PC-03 · EMULSIFIERS & THICKENERS · PERSONAL CARE

HS Stearic Acid

Stearic Acid

INCI	CAS	EC	FAMILY	SUPPLIED AS
Stearic Acid	57-11-4	200-313-4	PC-03	Flakes / beads

01 POSITIONING

The invisible architecture of every cream on the shelf.

02 HOW IT WORKS

01 **Crystalline scaffold**
Fatty chains crystallize into a lamellar network that gives creams their body and stability.

02 **Co-emulsification**
GMS reinforces the interfacial film and narrows droplet distribution.

03 **Melting-point design**
Stearic-based systems set sticks and balms at skin-melt temperatures.

03 APPLICATIONS

→ Creams and lotions
→ Sticks, balms and deodorants
→ Hair conditioners (body)
→ Soap superfatting
→ Candles and technical uses

04 FORMULATION GUIDE

USE LEVEL	1–8%
PH & CONDITIONS	Stable pH 4–9 in emulsified systems

— Melt into the oil phase at 70–75 °C; homogenize before cooling through the crystallization window.

— Combine GMS with stearic acid for the classic self-bodying pair.

05 TYPICAL SPECIFICATION

Appearance	White flakes / beads
Purity / monoester	per grade (GMS ≥ 95% mono)
Melting point	54–70 °C
Acid value	per TDS

Typical parameters. The specification on the current TDS and the per-lot Certificate of Analysis govern.

06 PERFORMANCE — DIRECTIONAL

Emulsion stability (cycle test, class data)
+ 2–4% fatty base

Body/viscosity per dosage
Fatty-base route

Values are representative of the ingredient class, based on published technical literature, and shown directionally for comparison. Final performance must be validated in the customer's formulation.